Growth of functions exercises

All questions are original. Unless stated otherwise n is a positive integer tending to infinity, comparison functions are eventually positive, and hidden constants do not depend on n.

03-E01 Prove bounds from quantifiers

For f(n)=5n²−9n+2, give explicit c_1,c_2,n_0 proving Θ(n²), then disprove O(n) against an arbitrary candidate constant. A numerical table is not a proof.

03-E02 Loose upper bounds and little-o

Let f=n² for even n and n for odd n, with g=n². Decide O, Ω, Θ, o, and ω, giving quantifier or subsequence arguments. Also give a function in Θ(g) whose ratio to g has no limit.

03-E03 General properties and cancellation

Prove $f + g = \Theta(\max(f, g))$ for eventually nonnegative f,g. Prove O transitivity and little-o transitivity. Do f=O(h) and g=O(h) imply f−g=Θ(h)? Give a cancellation example.

03-E04 Input quantifiers

For every size n, one input class costs n and another costs n². Give best-case and worst-case Θ bounds and the tightest uniform O and Ω bounds. Can any single g give a common Θ(g) bound over all inputs? Prove your answer.

03-E05 Integer splits and negative mod

For n=11 and interval p=7,r=17, find merge sort's midpoint and child lengths. Compute floor(−11/4), ceil(−11/4), and −11 mod 4. For x≥0 and positive integers a,b, prove $\lfloor \lfloor x/a \rfloor / b \rfloor = \lfloor x/(ab) \rfloor$.

03-E06 Logarithm and exponent traps

Order $(\lg n)^7, n^{0.1}, n \lg n, n^2, 3^n, n!$, proving little-o for adjacent terms. Is $2^{\lg^2 n}$ polynomially bounded? Are 3^{n+2} and 3^{2n} each Θ(3ⁿ)?

03-E07 General factorial bounds

Without Stirling prove lg(n!)=Θ(n log n), n!=o(nⁿ), and n!=ω(aⁿ) for every fixed a>0. Then use Stirling's error bounds to bracket the value for n=5. Distinguish relative and absolute error.

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Growth of functions exercises (continued)

03-E08 Iteration and stopping

List the full log-star chain for 256. Find $\lg^{(2)} 256$ and $(\lg 256)^2$. Give log-star pseudocode for every real n>0, prove that no nonpositive logarithm argument occurs, and prove termination. What representation problems need additional care in floating point?

03-E09 Fibonacci closed form and representation cost

Prove Binet's formula from $\phi^2=\phi+1$, $\psi^2=\psi+1$, and the base cases. Compute F_8 and prove $F_n=\Theta(\phi^n)$. Why does an O(1)-sized real expression not yield an O(1) word-RAM exact algorithm?

03-E10 Anonymous functions and uniform constants

Explain the quantifier direction of $6n^2 + \Theta(n) = \Theta(n^2)$. If h is eventually nonnegative and h(i)=O(i), prove $\sum_{i=1}^{n} h(i) = O(n^2)$. Why does $h_n(i) = ni$ not admit the same proof?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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